

AMIS 1012

Ethics in Computing



Chapter 3: Organisation Privacy (Part I)



Before we go into Organisation Privacy, let's get back to basics.

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- The future of **information and communications technology (ICT)** promises exciting challenges and opportunities.
- Powerful software and hardware systems are **changing** the way people and organizations interact in their daily life and on the Internet.
- To effectively and efficiently use computers and digital technology, you need to know the parts of an **information system**: people, procedures, software, hardware, data, and the Internet.

ICT



Parts of Information Systems	Description
People	End users using information systems
Procedures	The rules or guidelines for people to follow when using software, hardware, and data.
Software	A program consists of the step-by-step instructions that tell the computer how to do its work. The purpose of software is to convert data (unprocessed facts) into information (processed facts) . For example, a payroll program would instruct the computer to take the number of hours you worked in a week (data) and multiply it by your pay rate (data) to determine how much you are paid for the week (information).
Hardware	The equipment that processes the data to create information. It includes smartphones, tablets, keyboards, mice, displays, system units, and other devices. Hardware is controlled by software .

ICT



Parts of Information Systems	Description
Data	The raw, unprocessed facts , including text, numbers, images, and sounds, are called data. Processed data yields information. Using the previous example of a payroll program, the data (number of hours worked and pay rate) is processed (multiplied) to yield information (weekly pay).
Internet	Allows computers to connect and share information . This connectivity greatly expands the capability and usefulness of information systems.

Refer to O’Leary, D. A., O’Leary, T. J., & O’Leary, L. I. (2023). Computing Essentials 2023: Making it Work for You. McGraw Hill LLC. (Chapter 1 Information Technology, the Internet, and You) to go through the parts of information systems in detail.

<https://tarc.idm.oclc.org/login?url=https://resolver.vitalsource.com/9781265617974>



Learning Objectives (Part I)

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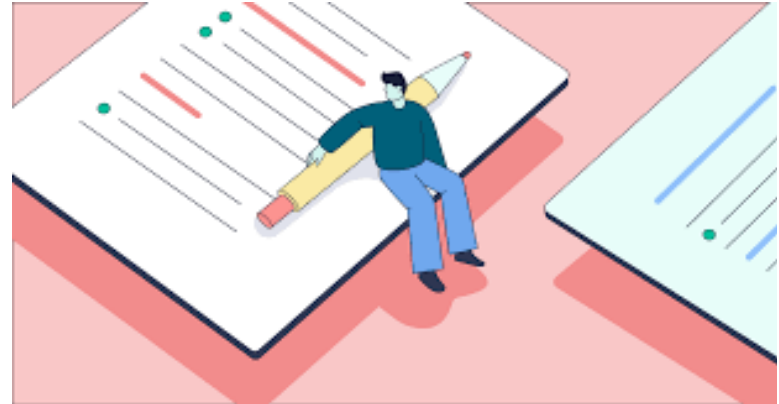
1. Explain the purpose of cyber security and its subdomains.
2. Discuss the various global and Malaysian cyber security and related laws.



Chapter Outline (Part I)

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1. Introduction to Cyber Security
2. Global Cyber Security and Related Laws (includes General Data Protection Regulation)
3. Malaysian Cyber Security Laws (includes Personal Data Protection Act and Cyber Security Act 2024)





1. Introduction to Cyber Security

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- Experts and policymakers have voiced **growing concern** in recent years about defending ICT networks from **cyber attacks**, which many experts predict will escalate in magnitude and intensity in the coming years.
- Cyber security refers to the **methods, techniques, and processes** used to **secure** the confidentiality, integrity, and functioning of information systems, networks, and data against cyber attacks or unauthorized access.



1. Introduction to Cyber Security

Cyber Security Subdomains	Description
Application Security	Developing protected application architectures, writing secure code, applying strong data input authentication to reduce the risk of any unwanted entry or alteration to application infrastructure.
Data protection and identity management	Identity protection refers to the mechanisms, procedures, and practices that allow legitimate persons to access information systems within an enterprise. Implementing strong information management systems to guarantee data integrity at rest and in transit is part of data protection.
Network Security	The implementation of both hardware and software frameworks to defend the network and facilities from unwanted access, delays, and misuse.
Mobile Security	Safeguarding all corporate and personal data stored on mobile devices such as cell phones, computers, tablets, and other similar devices from risks such as unauthorized entry, computer failure or theft, ransomware, and so on.



1. Introduction to Cyber Security

Cyber Security Subdomains	Description
Cloud Security	Development of stable and secure cloud architectures and software for businesses that use AWS, Google, Azure, and other cloud service providers.
Disaster Recovery and Business Continuity Plans (DR&BC)	DR&BC works with procedures, reporting, warnings, and preparations to help businesses prepare for maintaining business-sensitive services online after and during a crisis, as well as resuming some activities and systems.
Security Awareness and Training	Formal instruction on information security issues is critical for increasing knowledge of industry best practices, corporate processes, and regulations, as well as tracking and reporting malicious activity.

2. Global Cyber Security and Related Laws

- Cyber Security Laws
 - **A set of regulations and legal frameworks established by governments and organisations** to safeguard digital information, computer systems, networks, and the overall cyber environment.
 - These laws are designed to achieve several essential purposes:
 - Protection of Sensitive Data
 - Prevention of Cybercrime
 - Data Privacy
 - National Security
 - Consumer Trust

(Source: <https://www.internetsafetystatistics.com/cybersecurity-laws-beginner/>)



2. Global Cyber Security and Related Laws

- United States of America (USA)

- Computer Fraud and Abuse Act (CFAA)

- It is a USA cybercrime law that protects federal computers, bank computers, and computers connected to the Internet from cybercrime such as hacking, online fraud.

Source: <https://crsreports.congress.gov/product/pdf/RL/97-1025>

In their present form, the seven paragraphs of subsection 1030(a) outlaw

- computer trespassing (e.g., hacking) in a government computer, 18 U.S.C. 1030(a)(3);
- computer trespassing (e.g., hacking) resulting in exposure to certain governmental, credit, financial, or computer-housed information, 18 U.S.C. 1030(a)(2);
- damaging a government computer, a bank computer, or a computer used in, or affecting, interstate or foreign commerce (e.g., a worm, computer virus, Trojan horse, time bomb, a denial of service attack, and other forms of cyber attack, cyber crime, or cyber terrorism), 18 U.S.C. 1030(a)(5);
- committing fraud an integral part of which involves unauthorized access to a government computer, a bank computer, or a computer used in, or affecting, interstate or foreign commerce, 18 U.S.C. 1030(a)(4);
- threatening to damage a government computer, a bank computer, or a computer used in, or affecting, interstate or foreign commerce, 18 U.S.C. 1030(a)(7);
- trafficking in passwords for a government computer, or when the trafficking affects interstate or foreign commerce, 18 U.S.C. 1030(a)(6); and
- accessing a computer to commit espionage, 18 U.S.C. 1030(a)(1).

2. Global Cyber Security and Related Laws

- United States of America (USA) (Cont.)
 - The Stored Communications Act (SCA)
 - It is a USA cyber security law that protects stored electronic communications and data or "data-at-rest".
 - Examples of "data-at-rest" include the following:
 - Emails stored in a database
 - Text messages stored in a database
 - Instant messages stored in a database
 - Data in cloud storage
 - Data on a hard drive

Source: <https://www.findlaw.com/criminal/criminal-charges/hacking-laws-and-punishments.html>



2. Global Cyber Security and Related Laws

Provisions of the SCA most relevant to the government's access to communications and records include:

- United States of America (USA) (Cont.)
 - The Stored Communications Act (SCA) (Cont.)

- § 2701, which is the SCA's general prohibition against obtaining, altering, or preventing access to electronic communications in storage by intentionally accessing communications without authorization. The prohibition has several exceptions, including circumstances when the conduct to access the communication is authorized by the government.
- § 2702, which prohibits providers of electronic communication services (or "ECS," including cell phone providers, email providers, or social media platforms) and remote computing services (or "RCS," such as cloud computing providers) to the public from knowingly divulging communications held in electronic storage to any person or entity. Similar to § 2701, there are statutory exceptions to § 2702. The Department of Justice has taken the position that § 2702's prohibition on voluntary disclosures does not apply to sharing aggregate, de-identified non-content data with the government so long as it does not identify or otherwise provide information about any particular subscriber or customer.
- § 2703, which identifies how government entities can compel providers to disclose electronic communication information through a court-issued warrant, a court order, or an administrative subpoena. While § 2703 provides different degrees of process for law enforcement to obtain different types of communications, it generally requires a warrant for new ECS communications content (held for 180 days or less) and less robust protection for older content and non-content information.
- § 2705, which sets out the process by which the government may obtain a non-disclosure order to delay notification to an individual or entity under investigation that the government has requested the disclosure of a communication or record pursuant to § 2703.

Source:

<https://crsreports.congress.gov/product/pdf/LSB/LSB10801>

2. Global Cyber Security and Related Laws

- United States of America (USA) (Cont.)
 - The Electronic Communications Privacy Act (ECPA)
 - The ECPA forbids intentional interception of electronic communications in transit. It primarily acts as a restriction on wiretaps and the interception of signals. This type of data is also known as "data-in-transit." It refers to data while it is in transit to its destination.
 - Examples of data in motion include the following:
 - Emails
 - Text messages
 - Phone calls
 - Data while it is being uploaded from a cell phone to cloud storage
 - Data transfers between a hard drive and a computer

Source: <https://www.findlaw.com/criminal/criminal-charges/hacking-laws-and-punishments.html>

2. Global Cyber Security and Related Laws

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 - United States of America (USA) (Cont.)
 - The Electronic Communications Privacy Act (ECPA) (Cont.)
 - The ECPA has three titles:
 - **Title I** prohibits **wiretaps** (with some exceptions). It also prohibits the government from introducing illegally obtained communications as evidence in a criminal case.
 - **Title II** is known as the **Stored Communications Act**
 - **Title III** requires the **government to obtain authorization** to install certain surveillance technology, such as trap and trace devices and pen registers.

Source: <https://www.findlaw.com/criminal/criminal-charges/hacking-laws-and-punishments.html>



2. Global Cyber Security and Related Laws

- United States of America (USA) (Cont.)
 - Health Insurance Portability and Accountability Act of 1996 (HIPAA)
 - Federal law enacted in the United States in 1996
 - These regulations establish the standards for protecting the **privacy and security** of certain health information, specifically, **individually identifiable health information**.
 - **The Privacy Rule** protects the **privacy** of personal health information, also known as **Protected Health Information (PHI)** such as name, address, social security number, date of birth, medical record, health insurance information. The rule applies to all forms of PHI, whether electronic, written, or oral.
 - **The Security Rule** is a subset of the Privacy Rule and relates specifically to Electronic Protected Health Information (ePHI). The Security Rule requires covered entities to **ensure the confidentiality, integrity, and security of ePHI**.

Source: <https://www.exabeam.com/explainers/hipaa-compliance/hipaa-security-7-steps-for-compliance-with-the-security-rule/>

2. Global Cyber Security and Related Laws

- United States of America (USA) (Cont.)
 - Health Insurance Portability and Accountability Act of 1996 (HIPAA) (Cont.)
 - Example of **covered entities** who are responsible to protect individually identifiable health information:
 - **Health Care Providers**: hospitals, clinics, nursing homes, pharmacies, and doctors
 - **Health Insurance Companies**
 - **Health Care Clearinghouses**, such as billing services, community health management information systems
 - **Business Associates (eg. contractors, consultants, vendors, and subcontractors)** who are the third-party service providers who perform functions or activities on behalf of, or provide services to, a covered entities.

Source: <https://www.exabeam.com/explainers/hipaa-compliance/hipaa-security-7-steps-for-compliance-with-the-security-rule/>

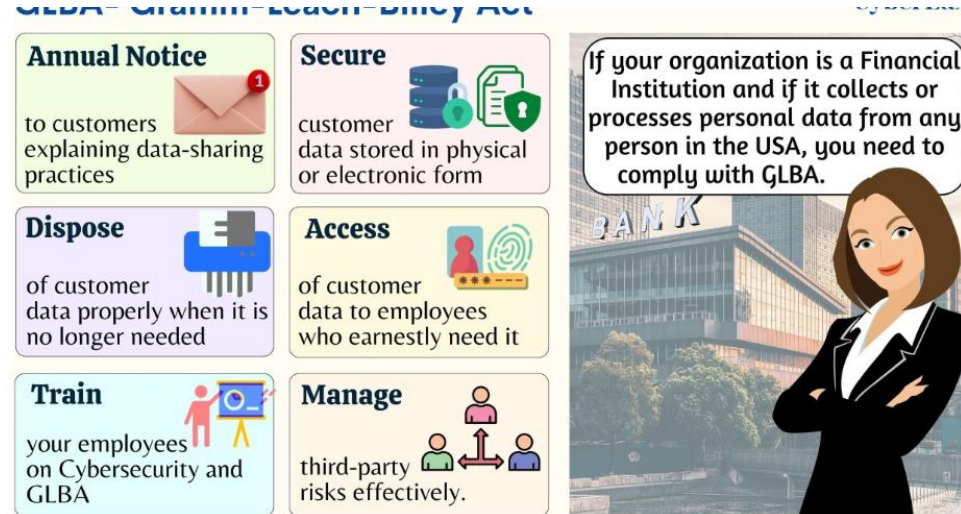
2. Global Cyber Security and Related Laws

- United States of America (USA) (Cont.)
 - California Consumer Privacy Act (CCPA)
 - State Level Privacy Law at USA.
 - Gives consumers more control over the personal information that businesses collect about them.
 - Secures privacy rights for California consumers, including:
 - **The right to know** about the personal information a business collects about them and how it is used and shared;
 - **The right to delete** personal information collected from them (with some exceptions);
 - **The right to opt-out** of the sale or sharing of their personal information; and
 - **The right to non-discrimination** for exercising their CCPA rights.

Source: <https://oag.ca.gov/privacy/ccpa>

2. Global Cyber Security and Related Laws

- United States of America (USA)
(Cont.)
 - Gramm-Leach-Bliley Act
 - requires financial institutions – companies that offer consumers financial products or services like loans, financial or investment advice, or insurance – to explain their information-sharing practices to their customers and to safeguard sensitive data.



Source: <https://www.linkedin.com/pulse/what-should-you-know-glba-gramm-leach-bliley-act-niharika-srivastav>

Source: <https://www.ftc.gov/business-guidance/privacy-security/gramm-leach-bliley-act>

2. Global Cyber Security and Related Laws

- Europe

- General Data Protection Regulation (GDPR)

Source: <https://gdpr.eu/what-is-gdpr/>

- **Toughest privacy and security law in the world**
- Though it was drafted and passed by the European Union (EU), **it imposes obligations onto organisations anywhere (affects every country)**, so long as they target or collect data related to people in the EU. Organisations need to prove **GDPR compliance**.
- The regulation was put into effect on May 25, 2018.
- The GDPR will **levy harsh fines** against those who violate its privacy and security standards, with penalties reaching into the tens of millions of euros.

2. Global Cyber Security and Related Laws

- Europe

- General Data Protection Regulation (GDPR) (Cont.)

Source: <https://gdpr.eu/what-is-gdpr/>

- **Personal data** — any information that **relates to an individual** who can be directly or indirectly identified. Example of personal data:
 - names
 - email addresses
 - location information
 - racial or ethnicity
 - Gender
 - biometric data
 - religious beliefs
 - web cookies
 - IP address
 - health and genetic data
 - political opinions



2. Global Cyber Security and Related Laws

- United Kingdom (UK)

- [Data Protection Act 2018](https://www.gov.uk/data-protection)

Source: <https://www.gov.uk/data-protection>

- UK's implementation of the General Data Protection Regulation (GDPR).

- China

- [Personal Information Protection Law \(PIPL\)](https://personalinformationprotectionlaw.com/)

Source: <https://personalinformationprotectionlaw.com/> and
<https://www.internetsafetystatistics.com/cybersecurity-laws-beginner/>

- aims to protect the personal information of Chinese citizens, regulate the cross-border transfer of data, and ensure that organisations handle data responsibly.

- India

- [Digital Personal Data Protection Bill, 2023](https://prsindia.org/billtrack/digital-personal-data-protection-bill-2023)

Source: <https://prsindia.org/billtrack/digital-personal-data-protection-bill-2023>

- apply to the processing of digital personal data within India where such data is collected online, or collected offline and is digitised.
 - also apply to such processing outside India, if it is for offering goods or services in India.



3. Malaysian Cyber Security Laws

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- The **National Cyber Security Agency (NACSA)** was officially established in February 2017 as the national lead agency for cyber security matters, with the objectives of securing and strengthening Malaysia's resilience in facing the threats of cyber attacks, by co-ordinating and consolidating the nation's best experts and resources in the field of cyber security. (Source: <https://www.nacsa.gov.my/>)



3. Malaysian Cyber Security Laws

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- [Malaysian Cyber Security Laws](#) as listed on National Cyber Security Agency (NACSA) official portal:
 - Copyright (Amendment) Act 1997
 - Computer Crimes Act 1997
 - Digital Signature Act 1997
 - Telemedicine Act 1997
 - Communications and Multimedia Act 1988
 - Electronic Commerce Act 2006



3. Malaysian Cyber Laws

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- [Malaysian Cyber Security Laws](#) as listed on National Cyber Security Agency (NACSA) official portal (Cont.):
 - Electronic Government Activities Act 2007
 - **Personal Data Protection Act 2010**
 - Penal Code
 - Anti-Fake News (Repeal) Act 2020
 - [Cyber Security Act 2024](#) (Act has been gazetted on 26 June 2024, **it has not yet been enforced.**)

Note: Go through the website for the brief description for all the listed Malaysian Cyber Security Laws



Question?

"Ethics is knowing the difference between what you have a right to do and what is right to do." - Potter Stewart



Disclaimer:

The slide content has been sourced from the following reference books and Internet resources. Refer to the following list for more details and further reading:

1. Awari, G.K., & Warjekar, S.V. (2022). *Ethics in Information Technology: A Practical Guide*. CRC Press, Taylor & Francis Group.
2. O'Leary, T., O'Leary, L., & O'Leary, D. (2024). Computing Essentials 2025: 2024 Release ISE. McGrawHill LLC.
<https://tarc.vitalsource.com/reader/books/9781264543182>
3. Internet Safety Statistics. (2024) Cybersecurity Laws: A Beginner's Guide
<https://www.internetsafetystatistics.com/cybersecurity-laws-beginner/>
4. Congressional Research Service. (2014). Cybercrime: An Overview of the Federal Computer Fraud and Abuse Statute and Related Federal Criminal Laws. <https://crsreports.congress.gov/product/pdf/RL/97-1025>
5. Congressional Research Service. (2022). Overview of Governmental Action Under the Stored Communications Act (SCA). <https://crsreports.congress.gov/product/pdf/LSB/LSB10801>
6. FindLaw. (2023). Hacking Laws and Punishments <https://www.findlaw.com/criminal/criminal-charges/hacking-laws-and-punishments.html>



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The slide content has been sourced from the following reference books and Internet resources. Refer to the following list for more details and further reading (Cont.):

7. Bureau of Justice Assistance, U.S. Department of Justice. Electronic Communications Privacy Act of 1986 (ECPA). <https://bja.ojp.gov/program/it/privacy-civil-liberties/authorities/statutes/1285#3-0>
8. Exabeam. HIPAA Security: 7 Steps for Compliance with the Security Rule. <https://www.exabeam.com/explainers/hipaa-compliance/hipaa-security-7-steps-for-compliance-with-the-security-rule/>
9. State of California Department of Justice. (2024). California Consumer Privacy Act (CCPA) <https://oag.ca.gov/privacy/ccpa>
10. Federal Trade Commission. The Gramm-Leach-Bliley Act. <https://www.ftc.gov/business-guidance/privacy-security/gramm-leach-bliley-act>
11. gdpr-info.eu. General Data Protection Regulation (GDPR). <https://gdpr-info.eu/>
12. UK Government. The Data Protection Act. <https://www.gov.uk/data-protection>
13. Personal Information Protection Law. <https://personalinformationprotectionlaw.com/>



Disclaimer:

The slide content has been sourced from the following reference books and Internet resources. Refer to the following list for more details and further reading (Cont.):

14. NACSA. (2024). MALAYSIAN CYBER LAWS. <https://www.nacsa.gov.my/legal.php>
15. PRS Legislative Research. (2024). The Digital Personal Data Protection Bill, 2023. <https://prsindia.org/billtrack/digital-personal-data-protection-bill-2023>
16. NACSA. (2024). MALAYSIA'S CYBER SECURITY ACT 2024 (ACT 854). <https://www.nacsa.gov.my/act854.php>

Please note that the slides' content is subject to changes and updates.